HU Collection
ESAT Mock Set F: Physics

Worked Solutions

Kevin Hu 29 October 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	F	E	D	C	E	C	B	B	A	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	A	F	D	C	D	A	C	F	B	D

Q	21	22	23	24	25	26	27			
Ans	B	B	F	E	A	A	E			

Each solution gives an intended route and names the trap the question was built around. Diagrams show the configuration each solution works with.

Question 1. Answer **F**.

The current, and so the coil's field, is greatest in size twice in every cycle (once in each direction) and reverses direction twice in every cycle. At 50 cycles per second that is 100 peaks and 100 reversals each second.

Soft iron is magnetised by induction, and its induced poles always arrange themselves so that it is attracted, whichever way the field points. So the force on the block in design X is always towards the coil, and it is greatest whenever the field is greatest in size: 100 times each second. I is true.

The permanent magnet in design Y keeps its own poles. When the end of the coil facing it is an unlike pole it is attracted; half a cycle later that end of the coil is a like pole and it is repelled. So the force reverses whenever the current reverses, 100 times each second, and III is true. But the strongest attraction happens only once per cycle (the strongest repulsion fills the other half), so it occurs 50 times each second, and II is false.

The usual slip is to treat both armatures alike: thinking the block is also pushed and pulled (giving III only), or counting cycles rather than peaks (giving II and III only).

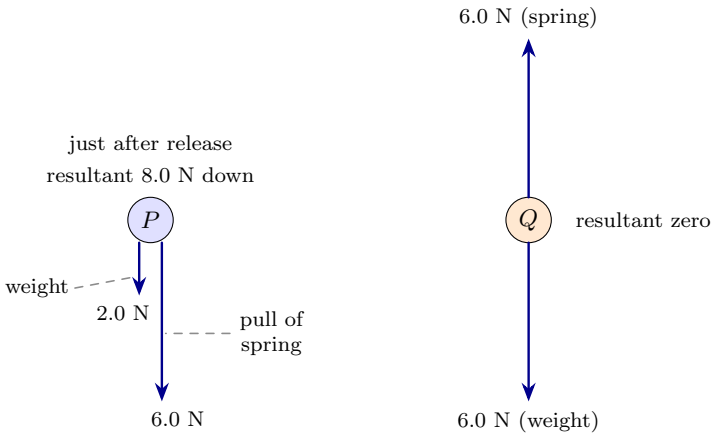
Question 2. Answer **E**.

A spring cannot change its length instantly, so at the moment of release it is stretched exactly as before and pulls with the same tension. Before release that tension held Q up, so it equals the weight of Q, 6.0 N.

Statement I is true. Q still feels 6.0 N up and 6.0 N down, so at that instant its resultant force, and its acceleration, is zero.

Statement II is false. P feels its own weight, 2.0 N, and the pull of the spring, 6.0 N, both downwards, so its acceleration is $8.0/0.20 = 40$ m s^{-2}. The value 30 m s^{-2} uses the spring force alone.

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Statement III is true. The spring pulls on P and on Q with equal and opposite forces, which cancel for the pair. The only external forces are the two weights, 8.0 N in total, so by $F = \Delta p/\Delta t$ the total momentum grows by 8.0 kg m s^{-1} each second, whatever the spring is doing.

The answer is I and III only. The picture of everything in free fall accelerating at g is true of the pair on average ($8.0/0.80 = 10$ m s^{-2}), but not of either ball on its own.

Question 3. Answer **D**.

Convert the trace into real times and voltages, change the sound, then convert back into divisions with the new settings.

On the upper screen one cycle takes 4 divisions, $4 \times 1.0 = 4.0$ ms, so the note is 250 Hz. The amplitude is 2 divisions, $2 \times 2.0 = 4.0$ mV.

Two octaves multiply the frequency by $2 \times 2 = 4$, giving 1000 Hz and a period of 1.0 ms. At 0.25 ms per division this is again 4 divisions. The amplitude becomes 2.0 mV, which at 0.50 mV per division is 4 divisions. So the new trace has the same horizontal spacing as before and twice the height: graph D.

The trap is reading the screen as a picture of the sound. Both settings were made four times more sensitive, so the higher, quieter note looks no more crowded and looks louder. Graph A shows the change in the sound with the old settings; graph E forgets the new time base and graph B forgets the new voltage scale; graph C treats two octaves as one doubling; graph F is the original trace.

Question 4. Answer **C**.

The loss to the surroundings carries on while the heater is on. So the rate of rise is set by the heater power minus the loss, and the rate of fall by the loss alone. Adding the two rates removes the unknown loss.

From the graph the water rises 0.40 K in 24 s and falls 0.40 K in 48 s. With heat capacity $C = mc$ and loss rate L,

$$\frac{630 - L}{C} = \frac{0.40}{24}\ \text{K s}^{-1}, \qquad \frac{L}{C} = \frac{0.40}{48}\ \text{K s}^{-1}.$$

Adding, $630/C = \frac{1}{60} + \frac{1}{120} = \frac{1}{40}$ K s^{-1}, so $C = 25\,200$ J K^{-1} and $m = 25\,200/4200 = 6.0$ kg.

As a check, the loss is $L = 25\,200/120 = 210$ W, and the heater is on for a third of each cycle, which supplies 210 W on average, as it must for the temperature to keep repeating.

Using the heating rate alone, as if nothing were lost while heating, gives 9.0 kg. Pairing 630 W with the cooling rate gives 18 kg. Averaging the two rates instead of adding them gives 12 kg. Supposing that 630 W for the whole 72 s cycle produces the 0.40 K rise gives 27 kg, and reading the swing as 0.80 K gives 3.0 kg.

Question 5. Answer **E**.

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Turn the reading into a volume, the volume into a mass of ice melted, and the mass into energy. The meniscus sweeps out a volume of $0.050\ \text{cm}^2 \times 20\ \text{cm} = 1.0\ \text{cm}^3$.

One gram of ice occupies $\frac{1}{0.90} = \frac{10}{9}\ \text{cm}^3$ and melts into $1\ \text{cm}^3$ of water, so every gram that melts shrinks the contents of the vessel by $\frac{1}{9}\ \text{cm}^3$. A shrinkage of $1.0\ \text{cm}^3$ therefore means that 9.0 g of ice melted, absorbing

$$0.0090 \times 3.3 \times 10^5 = 2970\ \text{J}.$$

This energy came from the metal cooling through $100\,^\circ\text{C}$, so

$$c = \frac{2970}{0.066 \times 100} = 450\ \text{J kg}^{-1}\,^\circ\text{C}^{-1}.$$

The contents shrink, so water is drawn back out of the tube: the meniscus moves towards the vessel.

Taking $1.0\ \text{cm}^3$ as a tenth of the volume of the water formed gives 10 g and 500; taking it as the volume of the ice that melted gives 0.90 g and 45; taking it as a ninth of that volume gives 8.1 g and 405; and reading $5.0\ \text{mm}^2$ as $0.50\ \text{cm}^2$ gives 4500.

Question 6. Answer **C**.

Charge is not the thing to follow here: each converter changes the p.d., so it changes the current too. Follow energy, using $E = QV$, and keep the units as mAh × V (milliwatt hours) throughout.

A full phone battery stores $3000 \times 3.6 = 10\,800$ in these units. The charging circuit is 80% efficient, so the cable must deliver $10\,800 \div 0.8 = 13\,500$, at 5.0 V. The charge through the cable is therefore $13\,500 \div 5.0 = 2700$ mAh, less than the 3000 mAh that goes into the phone's battery, because the cable runs at the higher p.d.

The power bank's cells hold $10\,000 \times 3.6 = 36\,000$. Its converter delivers 75% of this, 27 000, to the cable, and $27\,000 \div 13\,500 = 2$ complete charges exactly.

Treating charge as if it passed unchanged through the converters gives 3000 mAh (or 3750 mAh once the 80% is included), and comparing 10 000 with 3000 directly gives 3 charges.

Question 7. Answer **B**.

A running motor is not a resistor, so $R = V/I$ does not give its coil resistance. Use energy per second instead: electrical power in = heating in the coil + rate of doing work on the load, that is

$$VI = I^2 r + mgv.$$

The first load fixes r: $12 \times 1.0 = 1.0^2\, r + 0.50 \times 10 \times 2.0$, so $12 = r + 10$ and $r = 2.0\,\Omega$.

For the heavier load, $12 \times 3.0 = 3.0^2 \times 2.0 + 1.5 \times 10 \times v$, so $36 = 18 + 15v$ and $v = 1.2\ \text{m s}^{-1}$.

When the drum is clamped no work is done on a load, so all of the electrical power heats the coil: $12I = 2.0I^2$, giving $I = 6.0$ A. A stalled motor draws its largest current, not zero.

A check: the part of the supply p.d. not needed to drive current through the coil, $12 - Ir$, is 10 V at $2.0\ \text{m s}^{-1}$ and 6 V at $1.2\ \text{m s}^{-1}$, in proportion to the speed, as the induced p.d. in a turning coil should be.

Assuming the efficiency stays the same gives $2.0\ \text{m s}^{-1}$; assuming the useful power stays the same gives $0.67\ \text{m s}^{-1}$; using 12 V ÷ 1.0 A as the coil resistance gives a clamped current of 1.0 A.

Question 8. Answer **B**.

The RSI gives the rise: $3.2 \times 0.25 = 0.80$ m. Using $v^2 = 2gh$ each way, she lands at $\sqrt{2 \times 10 \times 0.45} = 3.0\ \text{m s}^{-1}$ downwards and leaves at $\sqrt{2 \times 10 \times 0.80} = 4.0\ \text{m s}^{-1}$ upwards.

Velocity is a vector. Taking upwards as positive, her momentum changes from -180 to $+240$ kg m s^{-1}, a change of 420 kg m s^{-1} in 0.25 s, so the resultant force on her is $420/0.25 = 1680$ N upwards.

The resultant is the push of the plate minus her weight, so the plate pushes with

$$1680 + 600 = 2280\ \text{N}.$$

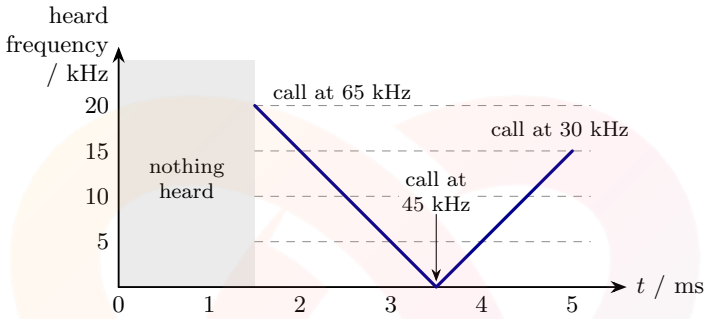
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Forgetting the weight gives 1680 N, and subtracting it gives 1080 N. Treating the change of velocity as $4.0 - 3.0$ gives 840 N (weight included). Using only the take-off speed gives 1560 N, and only the landing speed gives 1320 N.

Question 9. Answer **A**.

Work out when the loudspeaker's output is audible. The call frequency falls by 10 kHz every millisecond, and the output frequency is the gap between the call frequency and 45 kHz. The listener hears it when this gap is at most 20 kHz, that is while the call frequency is between 25 kHz and 65 kHz.

The call falls to 65 kHz at $t = 1.5$ ms, and it is still inside the band when it ends at 30 kHz. So the listener hears a sound for $5.0 - 1.5 = 3.5$ ms. The pitch she hears falls from 20 kHz to zero at $t = 3.5$ ms, when the call passes 45 kHz, and then rises again, reaching $45 - 30 = 15$ kHz at the end of the call.



Ignoring the limit of hearing gives the whole 5.0 ms, and 35 kHz is the inaudible output at the start of the call. Allowing only call frequencies above f_0 gives 2.0 ms and silence at the end, while counting only the part after the call passes f_0 gives 1.5 ms. Using the whole band from 25 to 65 kHz without noticing that the call stops at 30 kHz gives 4.0 ms.

Question 10. Answer **C**.

Count the charge per round trip. The ball arrives at the positive plate carrying $-q$ and leaves carrying $+q$, so the plate gives it $2q$; the supply replaces this through the ammeter. That happens once per round trip, and there are 5.0 round trips each second (one for each touch of the negative plate). So

$$I = 2q \times 5.0\ \text{s}^{-1}, \qquad q = \frac{40\ \text{nA}}{10\ \text{s}^{-1}} = 4.0\ \text{nC}.$$

On each crossing, a charge of size q moves through the full p.d. of the supply (positive charge from the positive plate to the negative plate, or negative charge the other way), so the field does work

$$W = qV = 4.0 \times 10^{-9} \times 4.0 \times 10^{3} = 1.6 \times 10^{-5}\ \text{J} = 16\,\mu\text{J}.$$

Check by power: the supply delivers $IV = 40\ \text{nA} \times 4.0\ \text{kV} = 160\,\mu\text{W}$, shared among 10 crossings each second, which is $16\,\mu$J per crossing.

Forgetting that each touch changes the charge by $2q$ gives 8.0 nC; using the change $2q$ as the charge carried across gives $32\,\mu$J.

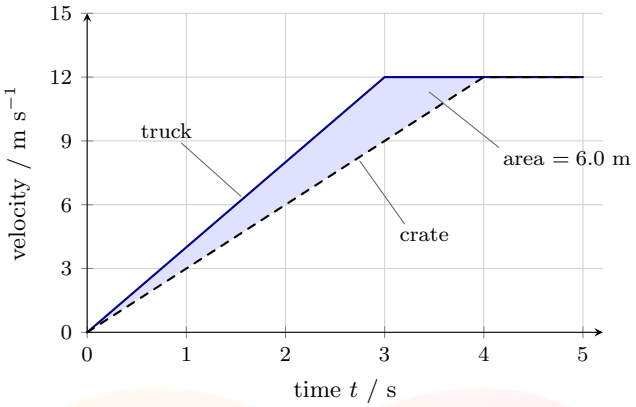
Question 11. Answer **A**.

To keep up with the truck the crate would need a forward friction force of $m \times 4.0$, but friction can supply at most $0.30 \times 10m = 3.0m$. So the crate slides backwards over the bed, while friction gives it a forward acceleration of 3.0 m s^{-2} relative to the road.

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The sliding does not stop at $t = 3.0$ s. The crate is still slower than the truck then, so it keeps sliding, and keeps accelerating at 3.0 m s^{-2}, until it reaches the truck's speed of 12 m s^{-1} at $t = 4.0$ s. After that no friction is needed and the crate rides at rest on the bed.

On a velocity-time graph, the distance the crate slides back along the bed is the area between the two graphs.



This is a triangle with vertices $(0, 0)$, $(3, 12)$ and $(4, 12)$: a base of 1.0 s along $v = 12$ and a height of 12 m s^{-1}, so the area is $\frac{1}{2} \times 1.0 \times 12 = 6.0$ m. (Check: by $t = 4.0$ s the truck has gone $18 + 12 = 30$ m and the crate $\frac{1}{2} \times 3.0 \times 4.0^2 = 24$ m.) So d must be at least 6.0 m.

Stopping at $t = 3.0$ s gives 4.5 m, and counting only the slide after $t = 3.0$ s gives 1.5 m. The values 12, 18 and 24 m are distances moved relative to the road, not along the bed.

Question 12. Answer **F**.

Write every quantity as a power of the scale factor 2. The muscle area, and so the push force F, scales as length squared: $\times 4$. The mass scales as volume: $\times 8$. The push distance s scales as length: $\times 2$.

Statement I is true. The work done in the push, Fs, scales by $4 \times 2 = 8$, the same factor as the mass. Since $\frac{1}{2}mv^2 = Fs$, the take-off speed is the same. (So both rise to the same height, $v^2/2g$: the model predicts that animals of very different sizes jump to similar heights.)

Statement II is true. The force is constant, so the acceleration is uniform and the mean speed during the push is $\frac{1}{2}v$, the same for both animals. Twice the distance at the same mean speed takes twice as long. (Equivalently, the acceleration F/m scales by $4/8 = \frac{1}{2}$, and the same speed gained at half the acceleration takes twice the time.)

Statement III is false. Mean power is work divided by time, which scales by $8/2 = 4$, not 8. Equivalently, power is force times mean speed, and only the force has changed, by a factor of 4.

The answer is I and II only. Choosing III comes from assuming that power scales with mass.

Question 13. Answer **D**.

Start at T. The tube is open to the atmosphere and holds only air, whose weight is ignored, so the pressure at its lower end is atmospheric pressure p_0. Bubbles are forming there, so the water at the level of T is also at p_0. Everything else follows from $p = \rho g h$, measured from this level.

I is true. The tap is 0.20 m below T, so the pressure there is $p_0 + 1000 \times 10 \times 0.20 = p_0 + 2.0$ kPa, however much water lies above T. This is why the bottle delivers water steadily.

II is true. The surface is 0.15 m above T, so the air above it is at $p_0 - 1000 \times 10 \times 0.15 = p_0 - 1.5$ kPa.

III is false, and is the trap. $pV =$ constant applies to a fixed mass of gas, but air is being added through the tube. As the surface falls, the column of water between the surface and T gets shorter, so the air pressure, which is p_0 minus the pressure due to that column, rises towards p_0.

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IV is true. With T lower, the tap is less far below the level at which the pressure is p_0, so the excess pressure at the tap, ρg times the height of T above the tap, is smaller.

The answer is I, II and IV only.

Question 14. Answer **C**.

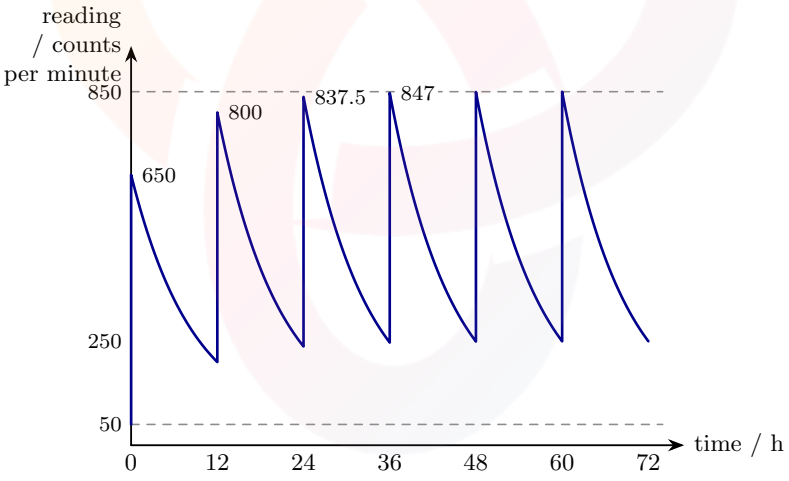
Use the steady state. Twelve hours is two half-lives, so between batches the activity in the tank falls to a quarter. Once the readings have settled, each batch must exactly replace what has decayed since the one before.

The tank's own count just after a batch is $850 - 50 = 800$ counts per minute, and it falls to $\frac{1}{4} \times 800 = 200$ before the next batch. So one batch supplies $800 - 200 = 600$ counts per minute, and just after the first batch, added to the empty tank, the reading was $600 + 50 = 650$ counts per minute.

For the second part, follow the shortfall below the settled value rather than the readings themselves. Just after a batch the tank gives C_n counts per minute, and $C_{n+1} = \frac{1}{4}C_n + 600$, while the settled value satisfies $800 = \frac{1}{4} \times 800 + 600$. Subtracting,

$$800 - C_{n+1} = \tfrac{1}{4}\,(800 - C_n)\,:$$

the shortfall also quarters every 12 hours. It is 200 after the first batch, then 50, 12.5, 3.1, …, so the readings are 650, 800, 837.5, 846.9, … The reading first exceeds 840 (shortfall below 10) just after the fourth batch.



Treating 12 hours as one half-life gives 450 and the seventh batch; forgetting to add back the background gives 600; and 250 is the settled reading just before a batch.

Question 15. Answer **D**.

Work in the yacht's frame, in which the transducers are at rest. If the air moves past the yacht at speed u from north to south, a pulse travelling south moves at $v + u$ relative to the yacht, and a pulse travelling north at $v - u$. So

$$v + u = \frac{0.168}{480 \times 10^{-6}} = 350 \text{ m s}^{-1}, \qquad v - u = \frac{0.168}{500 \times 10^{-6}} = 336 \text{ m s}^{-1}.$$

Half the sum and half the difference give $v = 343$ m s^{-1} and $u = 7.0$ m s^{-1}. The faster pulse travelled south, so relative to the yacht the air moves south at 7.0 m s^{-1}.

The temperature follows from $343 = 331 + 0.60\,\theta$: $\theta = 20\,^{\circ}\text{C}$. Combining the two reciprocal times separates the effect of the moving air from the effect of temperature, which is why real ultrasonic anemometers need no separate thermometer.

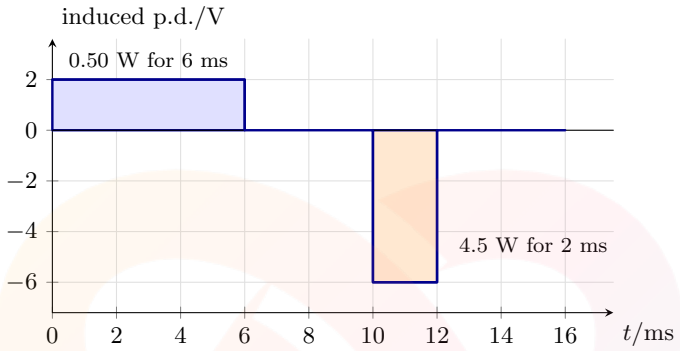
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The instrument measures the air's motion relative to the yacht. Relative to the water, the air's velocity is the yacht's velocity plus this: 6.0 m s^{-1} north plus 7.0 m s^{-1} south, which is 1.0 m s^{-1} towards the south.

Reporting 7.0 m s^{-1} ignores the yacht's motion, and 13 m s^{-1} adds the two speeds. Using 350 or 336 m s^{-1} on its own as the speed of sound gives 32 °C or 8 °C.

Question 16. Answer **A**.

Read the gradient of the graph, not its height. While I rises from 0 to 3 A in 6 ms it changes at 0.5 A per ms, so the induced p.d. is 2.0 V. While I falls from 3 A to 0 in 2 ms it changes at 1.5 A per ms, giving 6.0 V the other way. While I is steady, at 3 A or at zero, there is no induced p.d. at all.



The power in the resistor is V^2/R, whatever the direction of the p.d.:

rise: $\dfrac{2.0^2}{8.0} = 0.50$ W for 6 ms, which is 3.0 mJ;

fall: $\dfrac{6.0^2}{8.0} = 4.5$ W for 2 ms, which is 9.0 mJ.

So 12 mJ is dissipated in every 16 ms, a mean power of 0.75 W.

The pattern to notice: for a change ΔI made in time Δt, the energy is proportional to $(\Delta I)^2/\Delta t$, so the same 3 A change made three times as fast releases three times the energy. This is why an ignition coil switches its current off suddenly.

Averaging the p.d. first (1.5 V mean size) gives 0.28 W; using the rise alone gives 0.19 W, and the fall alone 0.56 W; leaving out the 8 ms with no change gives 1.5 W.

Question 17. Answer **C**.

Power is force times velocity, $P = Fv$, and neither the force nor the speed is given. The jump in the graph at $t = 1.0$ s separates them: the speed is the same just before and just after the jump, so the ratio of the powers is the ratio of the forces.

At constant speed the cable force equals the weight, mg; while the load speeds up it is $m(g + a)$. So

$$\frac{g+a}{g} = \frac{28.8}{24.0} = 1.2 \quad \Rightarrow \quad a = 2.0 \text{ m s}^{-2}.$$

Starting from rest, after 1.0 s the speed is 2.0 m s^{-1}. Then $mg = P/v = 24\,000/2.0 = 12\,000$ N, so $m = 1200$ kg. (The drop at $t = 6.0$ s to 19.2 kW, a factor of 0.8, means a deceleration of 2.0 m s^{-2}, which stops the load at $t = 7.0$ s, as the graph shows.)

For the height, the load starts and ends at rest, so all the work done by the cable becomes gravitational potential energy. That work is the area under the power-time graph:

$$\tfrac{1}{2}(1.0)(28.8) + (5.0)(24.0) + \tfrac{1}{2}(1.0)(19.2) = 144 \text{ kJ},$$

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so $h = 144\,000/12\,000 = 12$ m. (Check by kinematics: $1.0 + 10 + 1.0 = 12$ m.)

Taking the peak power to be mgv gives 1440 kg, and then 10 m. Forgetting to divide by the speed gives $mg = 24\,000$ N, so 2400 kg and 6.0 m. Multiplying 2.0 m s^{-1} by the whole 7.0 s gives 14 m, and counting only the steady stage gives 10 m.

Question 18. Answer **F**.

No external force acts, so the total momentum of the module and the astronauts stays zero at every instant, however the swap is done:

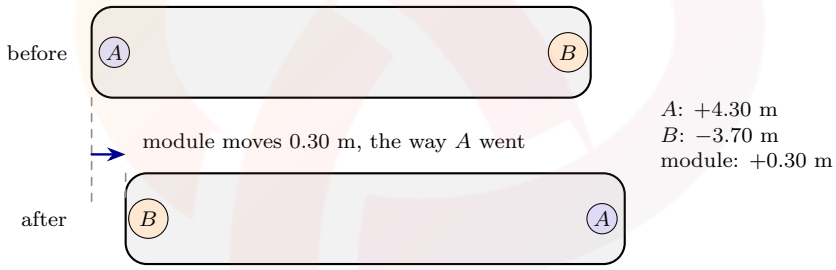
$$45v_A + 75v_B + 280v_M = 0.$$

The area under a velocity-time graph is a displacement, so the same combination of displacements, measured relative to space, is also zero: $45\,\Delta x_A + 75\,\Delta x_B + 280\,\Delta x_M = 0$. There is no need to know how the astronauts moved.

Take the direction in which A moves as positive, and let the module move D. Relative to the module A moves $+4.0$ m and B moves -4.0 m, so relative to space they move $4.0 + D$ and $-4.0 + D$:

$$45(4.0 + D) + 75(-4.0 + D) + 280D = 0 \quad \Rightarrow \quad 400D = 120, \quad D = +0.30 \text{ m}.$$

The heavier astronaut went backwards, so the module moves the way the lighter one went.



Treating the 4.0 m as displacements relative to space gives $120/280 = 0.43$ m. Leaving out the momentum of the module itself gives $120/120 = 1.0$ m. Using 8.0 m, the change in the astronauts' separation, as each one's movement gives 0.60 m. No external force means that the centre of mass of the whole system stays still, not that the module does.

Question 19. Answer **B**.

It is tempting to keep the outlet at 40°C (so the output halves) or to keep the output unchanged. Neither is consistent: slower water cools further, which lowers the mean temperature, which lowers the output. Write down both rules and solve them together.

Originally the mean temperature is 50°C, 30 K above the room, so the radiator gives out $1500/30 = 50$ W for each kelvin of excess. The water cools by 20 K while supplying 1500 W, so the flow delivers 75 W for each kelvin it cools; after halving the flow, 37.5 W K^{-1}.

Let the new outlet temperature be T. The water supplies $37.5(60 - T)$, and the radiator gives out $50\left(\frac{60+T}{2} - 20\right) = 25(T + 20)$. Setting these equal,

$$2250 - 37.5T = 25T + 500 \quad \Rightarrow \quad T = 28°\text{C},$$

and the output is $37.5 \times (60 - 28) = 1200$ W. Halving the flow cuts the output by only a fifth.

Keeping the outlet at 40°C gives 750 W, and keeping the output gives 1500 W. Assuming that the temperature drop doubles to 40 K and then using the radiator rule gives 1000 W. Finding $T = 28$°C but then using the original flow gives 2400 W. Measuring the temperatures from 0°C instead of from room temperature gives about 1290 W.

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Question 20. Answer **D**.

There are three unknown resistances and only two readings, so solving for all three is hopeless; but only ratios are asked for. Work with the current each part would carry per volt across it: write $a = 1/R_X$, $b = 1/R_Y$ and $c = 1/R_V$.

Voltmeter across X, reading V_1: the current through Y, which is $(12 - V_1)b$, splits between X and the voltmeter, so $(12 - V_1)b = V_1(a + c)$, giving

$$V_1 = \frac{12b}{a + b + c}.$$

In the same way, with the voltmeter across Y,

$$V_2 = \frac{12a}{a + b + c}.$$

The denominators are identical, so $V_1 : V_2 = b : a = R_X : R_Y$. The loaded readings are in exactly the ratio of the resistances, even though each is too low. Hence $R_X : R_Y = 4 : 6$, and without the voltmeter X takes $\frac{2}{5}$ of 12 V, which is 4.8 V.

For the voltmeter: $b = \frac{2}{3}a$, and from V_1, $a + b + c = 12b/4.0 = 3b = 2a$. So $c = 2a - a - \frac{2}{3}a = \frac{1}{3}a$, and $R_V = 3R_X$.

A check: with the voltmeter across X, the current through Y is $8.0/(1.5R_X)$, which equals $4.0/R_X + 4.0/(3R_X)$, as it should. Answering 2 gives the voltmeter's resistance as a multiple of R_Y, not R_X; sharing the missing 2 V equally between the readings gives 5.0 V.

Question 21. Answer **B**.

Measure every distance in seconds of sound travel, and convert at the end by multiplying by 340 m s^{-1}. The time of the stroke is unknown, so work with what does not depend on it.

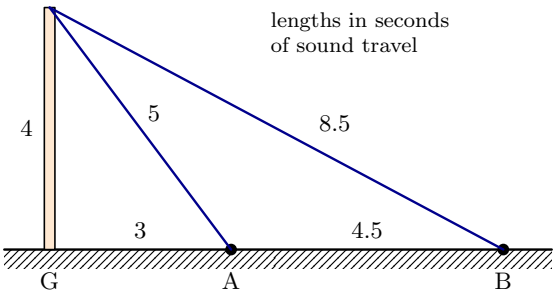
For a microphone a distance d from G, the thunder starts with sound from G, the nearest point of the channel, and ends with sound from the top, a distance $\sqrt{d^2 + H^2}$ away. So the thunder lasts $\Delta = \sqrt{d^2 + H^2} - d$, and the difference of two squares turns this into

$$H^2 = \left(\sqrt{d^2 + H^2} - d\right)\left(\sqrt{d^2 + H^2} + d\right) = \Delta\,(2d + \Delta).$$

The thunder starts 4.5 s later at B than at A, so if A is a seconds of travel from G, then B is $a + 4.5$. With $\Delta = 2.0$ at A and $\Delta = 1.0$ at B,

$$H^2 = 2.0\,(2a + 2.0) = 1.0\,(2a + 9.0 + 1.0),$$

so $4a + 4 = 2a + 10$, giving $a = 3.0$ and $H^2 = 16$. The channel is 4.0 seconds of travel high: $H = 4.0 \times 340 = 1360$ m $= 1.36$ km.



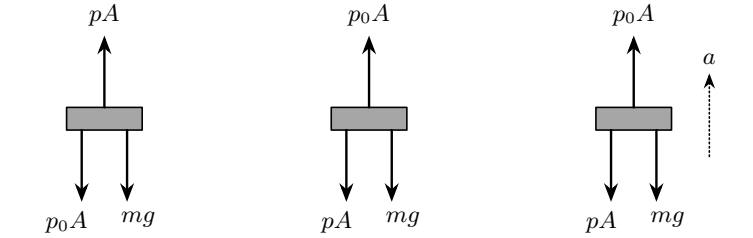
Check: the stroke happened at $t = 7.0$ s. A is 3.0 and B is 7.5 seconds of travel from G, and the right-angled triangles with sides 3, 4, 5 and 7.5, 4, 8.5 give the end times 12.0 s and 15.5 s.

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Assuming the stroke happened when the logger was switched on gives about 2.26 km from A's readings, which do not then fit B's. 0.68 km is just 2.0 s of travel, 1.70 km is the distance from A to the top of the channel, 1.02 km and 2.55 km are the distances of A and B from G, and 1.53 km is the distance between the microphones.

Question 22. Answer **B**.

Atmospheric pressure, the mass of the piston and its area are not given, and none of them is needed. Let p_0 be atmospheric pressure, m the mass of the piston and A its area, and let $x = mg/A$, the extra pressure that the weight of the piston alone could produce.



Open end up, at rest: the trapped air holds up the piston as well as the atmosphere pressing on it, so the air pressure is $p_0 + x$. Open end down, at rest: the atmosphere below pushes the piston up and must support both its weight and the push of the trapped air, so the air pressure is $p_0 - x$. By $pV =$ constant, with the volume proportional to the length,

$$(p_0 + x) \times 30 = (p_0 - x) \times 50,$$

so $80x = 20p_0$ and $x = \frac{1}{4}p_0$. For this air, $pL = 1.25\,p_0 \times 30 = 37.5\,p_0$ with L in centimetres.

In the lift the piston accelerates upwards at 5.0 m s$^{-2} = \frac{1}{2}g$. Taking upwards as positive, Newton's second law for the piston gives

$$p_0A - pA - mg = m \times \tfrac{1}{2}g,$$

so $p = p_0 - \frac{3}{2}x = p_0 - \frac{3}{8}p_0 = \frac{5}{8}p_0$. Then

$$L = \frac{37.5\,p_0}{\frac{5}{8}p_0} = 60 \text{ cm}.$$

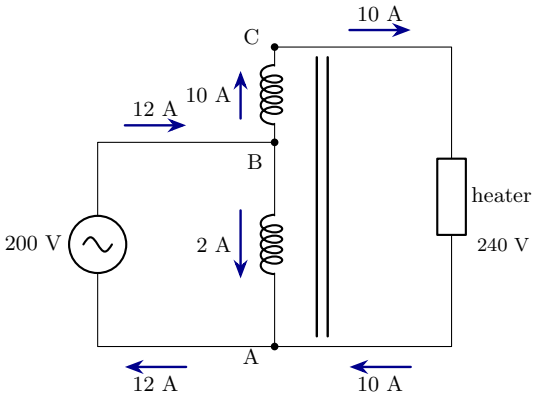
The upward acceleration has the same effect as a heavier piston, so the column stretches beyond its 50 cm length at rest. Applying the acceleration the wrong way gives 42.9 cm; 37.5 cm is the length when the piston's weight has no effect, with the cylinder horizontal or falling freely; 27.3 cm is the length with the open end upwards in the accelerating lift. Assuming the horizontal length is the mean of 30 and 50 cm leads to $x = \frac{1}{3}p_0$ and 80 cm, and treating the lift as doubling the effect of the piston's weight gives 75 cm.

Question 23. Answer **F**.

First the p.d.s. The 500 turns of AB have 200 V across them, so the 600 turns of AC have 240 V, and the heater current is $240 \div 24 = 10$ A. The heater takes 2400 W, so the supply current is $2400 \div 200 = 12$ A.

Now the key step: the coil is part of the circuit, so its sections carry currents decided by the junction rule. Section BC is in series with the heater, so it carries the heater's 10 A. At B the supply's 12 A arrives and 10 A of it leaves along BC, so only $12 - 10 = 2$ A flows through section AB, back to A.

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The rate of heating is I^2R with R proportional to the number of turns, so

$$\text{AB} : \text{BC} = 500 \times 2^2 : 100 \times 10^2 = 2000 : 10\,000 = 1 : 5.$$

This is the point of such a transformer: the long shared section carries only the small difference of the two currents, so it can be wound from thin wire.

Putting the supply's 12 A (or the heater's 10 A) through both sections gives 5 : 1; putting 12 A through AB and 10 A through BC gives 36 : 5; forgetting that resistance depends on the number of turns gives 1 : 25.

Question 24. Answer **E**.

Two ideas do all the work.

First, the energy dissipated. The resistive force F does negative work on the dart and positive work on the block. The dart moves further than the block by exactly the length ℓ of foam it passes through, so the kinetic energy lost overall is $F\ell$, whether or not the block moves. In Test 1 the block stays still and the dart only just reaches the far face, so $F\ell = \frac{1}{2}(0.090)(8.0)^2$.

Second, the threshold. In Test 2 the dart only just gets through if it reaches the far face at the moment its velocity becomes equal to the block's. Work in a frame moving with the block's initial velocity, in which the block starts at rest. If the dart's speed relative to the block is u, momentum gives a common velocity of $\frac{0.090}{0.250}u = 0.36u$, so the kinetic energy lost is

$$\tfrac{1}{2}(0.090)u^2 - \tfrac{1}{2}(0.250)(0.36u)^2 = (1 - 0.36) \times \tfrac{1}{2}(0.090)u^2.$$

Setting this equal to $F\ell$ gives $0.64u^2 = 8.0^2$, so $u = 10$ m s^{-1}.

That is the dart's speed relative to the block. The block is approaching at 3.0 m s^{-1}, so relative to the ground the dart needs only $10 - 3.0 = 7.0$ m s^{-1}.

Ignoring the block's motion gives 10 m s^{-1}, and adding its speed instead of subtracting gives 13 m s^{-1}. Applying the 3.0 m s^{-1} to the clamped result gives 5.0 or 11 m s^{-1}, and keeping the clamped result gives 8.0 m s^{-1}. Forgetting the square root ($8.0 \times \frac{25}{16} = 12.5$) and then subtracting 3.0 gives 9.5 m s^{-1}.

Question 25. Answer **A**.

Count gas particles. From radon-222 to lead-210 the mass number falls by 12. Only alpha decay changes the mass number, by 4 each time, so every radon atom that decays produces $12/4 = 3$ alpha particles. (The three alpha decays take the proton number from 86 down to 80, and two beta-minus decays bring it back up to 82; beta particles do not become gas.) So the decay of one radon atom removes one gas atom and adds three helium atoms: a net gain of two gas particles.

Let the radon initially contribute a pressure R. After one half-life, half the radon has decayed, so the pressure has risen by $2R \times \frac{1}{2} = R$. The rise is 20 kPa, so $R = 20$ kPa, one fifth of 100 kPa: one particle in five was a radon atom.

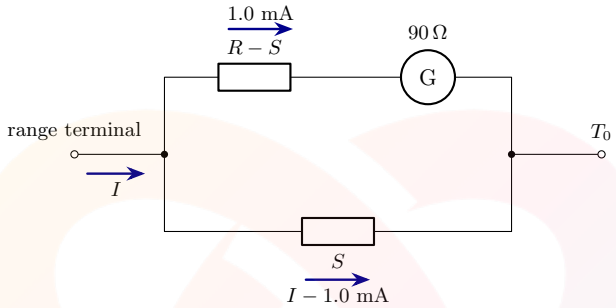
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After 11.4 days, three half-lives, $\frac{7}{8}$ of the radon has decayed and the pressure has risen by $2R \times \frac{7}{8} =$ 35 kPa, to 135 kPa. Notice that the rise is always proportional to the fraction of the radon that has decayed, so the later pressure follows from the first reading alone; the count of alpha particles is needed only for the fraction.

Counting the beta particles as gas particles gives a net gain of four per decay and a fraction of $\frac{1}{10}$, and forgetting that the radon atom itself disappears gives a gain of three and $\frac{2}{15}$; a gain of one gives $\frac{2}{5}$. Continuing the 20 kPa rise for every 3.8 days gives 160 kPa, 130 kPa is the pressure after only two half-lives, and 140 kPa is the pressure once all the radon has gone.

Question 26. Answer **A**.

Whichever terminal is used, the current splits into two paths that share the same p.d. One path is the part of the chain between that terminal and T_0; call its resistance S. The other is the rest of the chain followed by the meter. Write $R = R_1 + R_2 + R_3$ for the whole chain.



At full scale the meter path carries 1.0 mA through a resistance $90 + R - S$, so the shunt part carries $I - 1.0$ mA, and

$$(1.0\text{ mA})(90 + R - S) = (I - 1.0\text{ mA})\,S \quad\Longrightarrow\quad (1.0\text{ mA})(90 + R) = I\,S.$$

The left side is the same for every range. So the part of the chain used as the shunt is inversely proportional to the range.

On the 10 mA range the whole chain is the shunt, $S = R$: $1.0(90 + R) = 10R$, so $R = 10\,\Omega$ and the fixed product is $(1.0\text{ mA})(100\,\Omega) = 0.10$ V.

On the 100 mA range, $R_1 + R_2 = 0.10\text{ V} \div 0.10\text{ A} = 1.0\,\Omega$. On the 1.0 A range, $R_1 = 0.10\,\Omega$. Hence $R_2 = 0.90\,\Omega$ (and $R_3 = 9.0\,\Omega$).

Treating each range as a separate shunt across the bare meter (90/999 and 90/99) ignores the rest of the chain in series with the meter and gives $0.82\,\Omega$; giving $R_1 + R_2$ instead of R_2 gives $1.0\,\Omega$.

Question 27. Answer **E**.

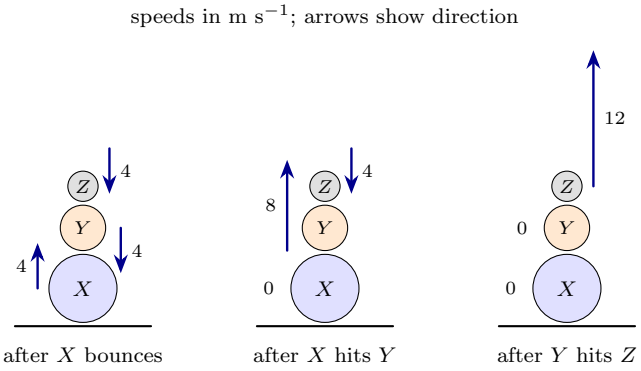
Solving each collision from scratch with momentum and kinetic energy means two simultaneous equations, one of them quadratic, twice over. Instead, guess an outcome and check it. For a head-on collision between two given bodies, the two conservation laws allow only one outcome (the other solution of the equations is the bodies passing through each other unchanged), so any outcome that conserves both momentum and kinetic energy is what happens.

Each ball reaches the floor at $\sqrt{2 \times 10 \times 0.80} = 4.0$ m s^{-1}. X rebounds at 4.0 m s^{-1} upwards and meets Y, which is still moving down at 4.0 m s^{-1}. Take upwards as positive.

X hits Y: try X stopping dead and Y leaving at $+8.0$ m s^{-1}. Momentum: $0.60(4.0) + 0.20(-4.0) = 1.6 = 0.20(8.0)$. Kinetic energy: $\frac{1}{2}(0.60)(16) + \frac{1}{2}(0.20)(16) = 6.4\text{ J} = \frac{1}{2}(0.20)(64)$. Both hold, so this is the outcome.

Y hits Z: Y at $+8.0$ m s^{-1} meets Z at -4.0 m s^{-1}. Try Y stopping and Z leaving at $+12$ m s^{-1}. Momentum: $0.20(8.0) + 0.10(-4.0) = 1.2 = 0.10(12)$. Kinetic energy: $\frac{1}{2}(0.20)(64) + \frac{1}{2}(0.10)(16) = 7.2\text{ J} = \frac{1}{2}(0.10)(144)$. Both hold again.

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Z leaves the floor at 12 m s^{-1}, three times its arrival speed, so it rises $3^2 \times 0.80 = 7.2$ m. All the kinetic energy of the stack, $\frac{1}{2}(0.90)(16) = 7.2$ J, ends up in the lightest ball and becomes $0.10 \times 10 \times 7.2 = 7.2$ J of potential energy.

Measuring from the release point gives 6.4 m. Supposing that each collision doubles the speed gives 16 m s^{-1} and 12.8 m. Using the result for very heavy lower balls, 28 m s^{-1}, gives 39.2 m. Dropping the $\frac{1}{2}$ in $v^2 = 2gh$ gives 14.4 m, a single doubling gives 3.2 m, and assuming nothing is gained gives 0.80 m.

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